



Banha-sasim-tang improves gastrointestinal function in loperamide-induced functional dyspepsia mouse model



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ABSTRACT

Ethnopharmacological relevance: *Banha-sasim-tang* (BST; *Hange-shashin-to* in Kampo medicine; *Banxia xiexin tang* in traditional Chinese medicine) is a traditional Chinese herbal medicine that has been commonly used for gastrointestinal disorders.

Aim of the study: To investigate the pharmacological effects of BST, a standardized herbal drug, on main symptoms of functional dyspepsia including delayed gastric emptying, and underlying mechanisms of action in mouse model.

Methods and materials: Balb/C mice were pretreated with BST (25, 50, 100 mg/kg, *po*) or mosapride (3 mg/kg, *po*) for 3 days, and then treated with loperamide (10 mg/kg, *ip*) after 19 h fasting. A solution of 0.05% phenol red (500 μ L) or 5% charcoal diet (200 μ L) was orally administered, followed by scarifying and assessment of gastric emptying or gastro-intestinal motility. C-kit (immunofluorescence), nNOS (western blot) and gastric contraction-related gene expression were examined in stomach tissue.

Results: The loperamide injection substantially delayed gastric emptying, while the BST pretreatment significantly attenuated this peristaltic dysfunction, as evidenced by the quantity of stomach-retained phenol red ($p < 0.05$ or 0.01) and stomach weight ($p < 0.05$ or 0.01). The BST pretreatment significantly tempered the loperamide-induced inactivation of c-kit and nNOS ($p < 0.05$ or 0.01) as well as the contraction-related gene expression, such as the 5HT₄ receptor (5HT₄R), anoctamin-1 (ANO1), ryanodine receptor 3 (RyR3) and smooth muscle myosin light chain kinase (smMLCK). The BST pretreatment also significantly attenuated the alterations in gastro-intestinal motility ($p < 0.01$).

Conclusion: Our results are the first evidence of the prokinetic agent effects of *Banha-sasim-tang* in a loperamide-induced FD animal model. The underlying mechanisms of action may involve the modulation of peristalsis via activation of the interstitial cells of Cajal and the smooth muscle cells in the stomach.

1. Introduction

Functional dyspepsia (FD) is one of the most common functional gastrointestinal (GI) disorders affecting 10–30% of the population worldwide (Mahadeva and Goh, 2006). FD is mainly characterized by symptoms such as postprandial fullness, early satiation, or epigastric pain/burning, without any organic problem to explain the symptoms

(Stanghellini et al., 2016). FD is known to impair the physical, mental and social aspects of health-related quality of life (HRQoL) more than other GI functional disorders, such as, irritable bowel syndrome (IBS) and gastroesophageal reflux disease (GERD) (El-Serag and Talley, 2003).

FD can be subdivided into postprandial distress syndrome (PDS) and epigastric pain syndrome (EPS) (Asano et al., 2016). PDS presents

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postprandial fullness and early satiety while EPS dose epigastric burning and epigastric pain, these two subtype are corresponding to 67.5% and 48.2% of FD patients (Carbone et al., 2015). Interestingly, the PDS subtype of FD is known to be prevalent in the Asian population (Asano et al., 2017). One electrogastrography (EGG)-derived clinical study found a close link between delayed gut motility and impairment of gastric emptying ability (Chen et al., 1996). Gut motility depends on peristalsis conducted by sequential contraction and relaxation of smooth muscle according to the slow wave produced by the interstitial cells of Cajal (ICC) (Sanders, 1996).

In general, the main causes of FD include *helicobacter pylori* infection, visceral hypersensitivity, acid disorders psychosocial factors and abnormal gut motility (Enck et al., 2017). Based on these etiologic factors, the prescribed treatments have been *H. pylori* eradication, histamine-type-2 receptor antagonists, proton pump inhibitors (PPI), selective serotonin reuptake inhibitors (SSRIs), tricyclic antidepressants (TCA) or prokinetic drugs (Tack and Camilleri, 2018). However, these therapeutics had the clinical limitations such as, lower response and frequent relapse after cessation (Lacy et al., 2012). Furthermore, they had unwanted side effects such as increased susceptibility to infections and secondary hypergastrinemia from long-term use of PPI and arrhythmia by from use of cisapride, 5HT₄ receptor agonist (De Maeyer et al., 2008; Shiotani et al., 2018).

On the other hand, many clinical trials have reported the beneficial effects of traditional Chinese medicine (TCM) on FD, especially with respect to QoL improvement from acupuncture (Jin et al., 2015; Kim et al., 2014), and the prokinetic effects of herbal formulae. *Banxia-sasim-tang* (BST) is a traditional Korean medicine (called as *Banxia xiexin tang* in traditional Chinese medicine and *Hange-shashin-to* in Kampo medicine). BST has been traditionally prescribed for patients suffering from various types of dyspepsia in Eastern Asian countries including China, Korea and Japan (Zhao et al., 2013). One clinical study reported the safety and efficacy of BST for FD symptoms (Park et al., 2010). BST has also showed protective effects against chronic acid reflux esophagitis (Xu, 2006) and regulation of gastric peristalsis and the lower esophageal sphincter in GERD mouse model (Shin et al., 2017). However, there is no experimental evidence of BST effects for FD, especially for GI motility, and nor of its underlying mechanisms of action.

In order to support the clinical relevance of BST, this study evaluated the pharmacological effects of BST on the main symptoms of FD including delayed gastric emptying, and investigated the underlying mechanisms of action in a loperamide-induced mouse model.

2. Materials and methods

2.1. Materials

The following reagents were obtained from Sigma-Aldrich (MO, USA); mosapride hydrochloride, phenol red, loperamide hydrochloride, trichloroacetic acid (TCA), carboxymethyl cellulose sodium salt (CMC-Na), gum arabic, charcoal, Tween 20, xylene, 4',6-diamidino-2-phenylindole dihydrochloride (DAPI), aqueous mounting buffer, sodium hydroxide and bovine serum albumin (BSA).

Other reagents were obtained from the following manufacturers: Triton X, paraformaldehyde, acetyl alcohol, H₂O₂, methylene alcohol, citrate buffer (Duksan Science, Seoul, Korea; Daejung Chemicals & Metals Co., Siheung, Korea; Junsei Chemical Co., Ltd., Tokyo, Japan), normal goat serum, Alexa 484, c-kit, nNOS, β -actin and secondary horseradish peroxidase (HRP)-conjugated antibodies (Abcam, MA, USA; Thermo-Fisher Scientific, PA, USA; Santa Cruz Biotechnology, CA, USA) and AMRESCO.

2.2. Preparation of BST syrup and fingerprinting

The BST syrup (17JW-08) was purchased from Jeong-woo Pharmaceutical Company Ltd., (Seoul, Korea). Briefly the eight herbal

materials were extracted by boiled water, and extracts were filtered, concentrated and lyophilized. Finally BST syrup is produced with four additives. The BST syrup consisted of the following herbs: water extract of *Pinelliae* tuber, *Scutellariae* radix, *Ginseng* radix, *Glycyrrhizae* radix, *Zingiberis* rhizome recens, *Coptidis* rhizome and *Zizyphi* fructus with additives that included β -cyclodextrin, apple concentrate, carboxymethyl cellulose and sodium benzoate, as set out in Supplementary Table 1. Voucher specimens were retained at the laboratory of Oriental medicine development, National Development Institute of Korean Medicine.

To verify the quality of the BST syrup, fingerprinting analysis was performed with six reference compounds (coptisine, baicalin, palmatine, berberine, baicalin and glycyrrhizic acid). The reference compound of each herb was selected based on the Korea herbal pharmacopoeia (Korea Food and Drug Administration, 2002). The BST and all reference compounds were analyzed using a high-performance liquid chromatography system (HPLC, Agilent 1100 system, CA, USA) equipped with an auto sampler (G11313A), column oven (GA1316A), binary pump (G1312), diode-array-detector (DAD), and degasser (GA1379A). The analytical column used was an YMC-Triart C18 (4.6 $\times$ 250 mm²; particle size, 5 μ m; YMC, Kyoto, Japan). The mobile phase consisted of 0.1% Phosphoric acid in H₂O (A) and 0.1% Phosphoric acid in Acetonitrile (B). The gradient flow was as follows: 0–40 min, 25–40% B; 40–60 min 40–65% B. The analysis was conducted at a flow rate of 1.0 mL/min and detection wavelength of 254 nm. The injection volume was 10 μ L and concentration was 0.1 g/mL. The data were acquired and processed by ChemStation software (Agilent Technologies, CA; Fig. 1A).

2.3. Animals and experimental design

Ninety-six specific pathogen-free male Balb/C mice (6 weeks old, 19–21 g) were obtained from a commercial animal breeder (Daehanbio-link, Chongbuk, Rep. of Korea). Mice were housed in an environmentally controlled room maintained at 22 $\pm$ 2 °C and 60 $\pm$ 5% relative humidity under a 12/12 h light/dark cycle. The mice were fed a commercial standard chow diet (Daehanbio-link) with tap water *ad libitum*.

After acclimation for 7 days, the mice were randomly allocated into three experimental sets; first set for gastric emptying test (n = 48, 8 mice per group), second set for small intestinal motility (n = 30, 5 mice per group) and third set for histopathology, western blot and qPCR (n = 18, 3 mice per group), respectively. Each set divided into six groups: naive, loperamide (10 mg/kg), three doses of BST (25, 50, 100 mg/kg) and mosapride (3 mg/kg, as a positive control). The clinical dose of BST is 3 g daily for human adult (about 60 kg) traditionally, which is equivalent to 615 mg/kg of mouse. However, our several pre-experiments found that even though low dose (50 mg/kg) showed the prokinetic action, which we decided our experimental dose. In addition, generally clinical dose of mosapride is 15 mg/60 kg adult, which is equivalent to 3.1 mg/kg of mouse (Reagan-Shaw et al., 2008). The BST and mosapride were dissolved in distilled water. Each group was orally administered distilled water (naive and loperamide group), BST or mosapride for 3 consecutive days (Ekor, 2014; Huang et al., 2004). The experimental schedule was designed as shown in Fig. 1B–C. This experimental design was decided based on our preliminary experimental results, regarding dose of loperamide (2, 5 and 10 mg/kg, single intraperitoneal injection), time points of phenol red administration and sacrifice of mice. Ten mg/kg of loperamide induced 50% inhibition of gastric emptying (Supplementary Fig. 1A).

Animal care and experiments were conducted in accordance with the guidelines issued by the Institutional Animal Care and Use Committee of Daejeon University (Daejeon, Republic of Korea; Approval No. DJUARB 2017-036) and the Guide for the Care and Use of Laboratory Animals published by the United States National Institutes of Health.

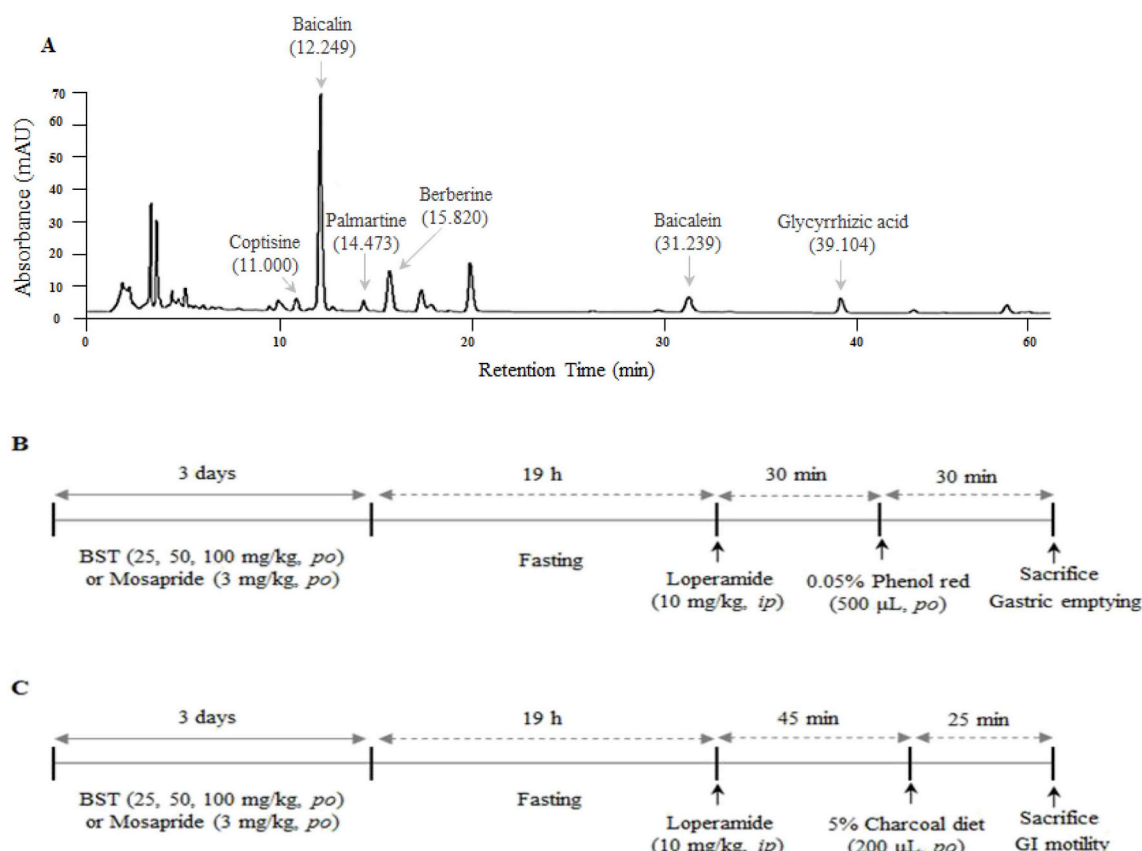


Fig. 1. Chemical composition analysis by HPLC and scheme of experiment. Composition of the *Banha-sasim-tang* (BST) was subjected to HPLC (A). The experimental schedule is summarized in (B) and (C).

2.4. Stomach weight and evaluation of gastric emptying using phenol red

The mice were fasted for 19 h with free access to tap water, based on the pilot experiment determined the appropriate fasting time (Supplementary Fig. 2A). The choice of phenol red solution volume (500 µL/mouse) and the time point of 50% delayed gastric emptying (30 min after intraperitoneal injection with loperamide 10 mg/kg) were made by our pilot experiments (Supplementary Figs. 2B–C) and other study protocol (Asano et al., 2016; Scarpignato et al., 1980; Shin et al., 2018; Lee et al., 2016). Accordingly, 30 min later of 0.05% phenol red administration, the mice were sacrificed, and then stomachs were immediately removed and weighted.

Next, to measure the absorbance of stomach-retained phenol red, the stomachs were homogenized in 5 mL of 0.1 N sodium hydroxide solutions and; 0.5 mL of trichloroacetic acid (20% w/v). The homogenates were centrifuged at 3000 rpm for 20 min; 1 mL of supernatant was added to 4 mL of 0.5 N sodium hydroxide solutions, and the absorbance of this pink colored liquid was measured at 560 nm using a spectrophotometer.

The gastric emptying value was calculated according to the following formula:

$$\text{Gastric emptying (\%)} = (1 - X/Y) \times 100.$$

X: Absorbance of phenol red solution remaining in the stomach.

Y: Absorbance of phenol red mixed with sodium hydroxide solution in test-tube conditions.

2.5. Evaluation of intestinal transit rate using charcoal diet

To determine the intestinal transit rate, we used charcoal diet method, as 5% charcoal dissolved in distilled water with gum Arabic, as described previously (Nunes Marona and Bastos Lucchesi, 2004). The

charcoal diet was orally administered (200 µL/mouse) 45 min after the loperamide intraperitoneal injection. The mice were sacrificed 25 min after charcoal diet administration, and the intestinal transit distances were determined by measuring the distance the charcoal had migrated through the small intestine (from the pylorus to the cecum). The above time points were determined by other study protocol (Bafar et al., 2017; Mittelstadt et al., 2005) and our pilot experiments; time of charcoal diet passage about 80% (25 min) under normal condition and 80% inhibition of GI motility (45 min) under loperamide injection (Supplementary Figs. 2D–E).

2.6. Immunofluorescence staining analysis in stomach tissue

Paraffin sections of stomach tissue (4 µm) were dried at 60 °C for 15 min. The sections were then subjected to deparaffinization with xylene and to rehydration with ethanol (with 100%, 95%, 85%, 70%, and 50% ethanol and tap water for, 3 min in each case). Antigens were retrieved in 10 mM sodium citrate buffer for 10 min. After washing three times, non-specific binding affinity was blocked for 1 h using normal goat serum, and then the slides were incubated with the anti-c-kit antibody (1:200) overnight at 4 °C. After washing, slides were incubated with goat anti-rabbit Alexa Fluor 488 conjugated secondary antibody (1:200) for 1 h at RT. After washing three times for 10 min, the slides then were incubated with DAPI (1 µg/mL) for 2 min at RT in the dark. The c-kit signal was observed using as Axio-phot microscope (Carl Zeiss, Germany). The c-kit protein expression was semiquantified using ImageJ (NIH).

2.7. Western blot analysis

To determine the stomach protein level of neuronal nitric oxide synthase (nNOS), stomach tissue was prepared in the RIPA lysis buffer.

The protein was denatured by boiling for 5 min, and was then separated by 10% polyacrylamide gel electrophoresis and transferred to polyvinylidene fluoride (PVDF) membranes. After blocking in 3% BSA for 1 h, the membranes were probed with primary antibodies such as nNOS (1:1000, ab76067) or β -actin (1:1000, MA5-11869) overnight at 4 °C. The membranes were washed and incubated with HRP-conjugated anti-rabbit (1:2500, against nNOS) or anti-mouse antibody (1:2500, against β -actin) for 40 min. The protein was visualized using an enhanced chemiluminescence (ECL) advanced kit (Thermo Fisher Scientific, USA). The protein expression was observed using as FUSION Solo System (Vilber Lourmat, France). The protein expression was semi-quantified using ImageJ (NIH).

2.8. Gene expression analysis by quantitative real-time PCR

To determine contraction-related gene expressions such as the 5HT₄ receptor (5HT₄R), anoctamin-1 (ANO1), ryanodine receptor 3 (RYR3) and smooth muscle cell myosin light chain kinase (smMLCK), total mRNA was extracted from stomach tissue using Trizol reagent (Molecular Research Center, Cincinnati, OH, USA). The cDNA was synthesized from total RNA (1 μ g) using a high-capacity cDNA reverse transcription kit (Ambion, USA). Real-time PCR was performed using SYBR Green PCR Master Mix (Applied Biosystems, USA) and primers as described in Table 1. Gene expression data were analyzed using the IQ5 PCR Thermal Cycler (Bio-Rad, USA).

2.9. Statistical analysis

All data are expressed as the mean $\pm$ standard deviation (SD). The statistical significance was analyzed by one-way analysis of variance (ANOVA), followed by post hoc multiple comparisons with Fisher's LSD *t*-test using IBM SPSS statistics software, ver. 20.0 (SPSS Inc., Chicago, IL, USA). The differences observed were considered statistically significant at $p < 0.05$ or 0.01 .

3. Results

3.1. Effects on delayed gastric emptying

Loperamide injection induced fullness of the forestomach, whereas BST pretreatment attenuated it, as notably seen by naked-eye observation (Fig. 2A). This finding was validated with the quantitative assays. The stomach weights in the BST-treated groups were significantly lower in comparison with the weights of the loperamide group ($p < 0.05$ or $p < 0.01$, Fig. 2B). BST pretreatment also significantly attenuated the retained amount of phenol red inside the stomach as compare with the loperamide group ($p < 0.05$ or $p < 0.01$, Fig. 2C). Mosapride pretreatment also showed similar effects to BST treatment.

Table 1
Summary for gene sequence.

Gene	Primer	Primer sequence (5'→3')
5HT ₄ R	Forward	ATG GTC AAC AAG CCC TAT GC
	Reverse	AGG AAG GCA CGT CTG AAA GA
ANO1	Forward	GGT GTC GGG TTT GTG AAG AT
	Reverse	TGCACGTTGTCTCTTCAGG
RYR3	Forward	GGC CAA GAA CAT CAG AGT GAC TAA
	Reverse	TCA CTT CTG CCC TGT CAG TTT C
smMLCK	Forward	AGA AGT CAA GGA GGT AAA GAA TGA TGT
	Reverse	CGG GTC GCT TTT CAT TGC
GAPDH	Forward	CAT GGC CTT CCG TGT TCC T
	Reverse	CCT GCT TCA CCA CCT TCT TGA

3.2. Effects on c-kit protein expression on stomach tissue

Loperamide injection notably reduced c-kit protein expression in stomach tissue (approximately 0.5-fold), whereas BST pretreatment significantly tempered this alteration in Immunofluorescence intensity ($p < 0.05$ or $p < 0.01$, Fig. 3A and B). Mosapride pretreatment also showed similar effects to BST treatment.

3.3. Effects on molecules associated with gastrointestinal motility

Loperamide injection drastically decreased nNOS protein expression in stomach tissue (approximately 0.5-fold), whereas BST pretreatment significantly increased nNOS protein expression in contrast with the loperamide group ($p < 0.01$, Fig. 4A and B). Loperamide injection also downregulated the smooth muscle contraction-related gene expressions such as 5HT₄R, ANO1, RYR3 and smMLCK. These alterations were attenuated by BST pretreatment ($p < 0.05$ or $p < 0.01$, mainly in BST 100 mg/kg, Fig. 4C). Mosapride exerted positive effects in the nNOS protein and 5-HT₄R gene expression only.

3.4. Effects on small intestinal motility

Loperamide injection significantly inhibited motility in the small intestine (approximately 0.25-fold) compared with the effect in the naive treatment group. This inhibited small intestinal motility was prominently recovered by pretreatment with BST ($p < 0.01$, Fig. 5A and B). Mosapride pretreatment also significantly improved small intestinal motility similarly to BST.

4. Discussion

To evaluate the pharmacological effects of BST and gain sight into its underlining mechanisms of action, we adopted an animal model of functional dyspepsia using loperamide injection. Loperamide, a μ -opioid receptor agonist, has been used to induce dyspeptic symptoms in rodent model (Lee et al., 2016). Gastric emptying is conducted by gastrointestinal motility, which depends on activations of autonomic neurons and interstitial cells of Cajal (ICC) (Farrugia, 2008; Sanders et al., 2010). Opioid agonists including morphine and loperamide, suppress acetylcholine release, thereby preventing both activation of muscarinic receptors on excitatory motor neurons and nitric oxide release from inhibitory motor neurons (Sanders et al., 2014). The μ -opioid receptor couples to the G₀ subtype of G protein which links the μ -opioid receptor to inhibition of Ca²⁺ channels and activation of K⁺ channels in ICC (Fujita et al., 2001; Shen and Surprenant, 1991). In recent studies, loperamide led to delayed gastric emptying and slowed GI motility in mouse model (Goineau et al., 2015; Nam et al., 2016).

As expected, a single dose of loperamide injection (10 mg/kg) delayed gastric emptying, evidenced by the observation of postprandial fullness and increase of stomach weight and retained phenol red inside stomach. Body weight and food and water intake were not affected in this model (Supplementary Figs. 1B–D). Pretreatment with BST significantly prevented this delay of gastric emptying (Fig. 2A–C). The delayed gastric emptying is a typical feature of FD in most of clinical studies, and especially for PDS subtype (Hafeez et al., 2018; Quarter et al., 1998). Compared with FD patients with the EPS subtype, FD patients with PDS who complained of postprandial fullness reported greater improvement in QoL after treatment (Aro et al., 2011). Clinical applications of loperamide (8 mg/kg, single dose) have prolonged gastric residence time up to 6.4-fold in healthy subjects (Kirby et al., 1989). In our model, GI motility was also drastically inhibited by loperamide (Fig. 5A and B). Previous studies have reported a high degree of overlap (18.5–19.4%) between irritable bowel syndrome (IBS) and functional dyspepsia (Futagami et al., 2013; Kaji et al., 2010). In particular, postprandial fullness is a main complain of patients with both constipation-predominant IBS and FD (Choi et al., 2017). The loperamide-

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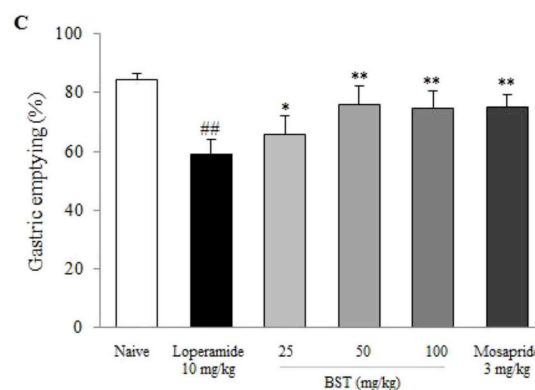
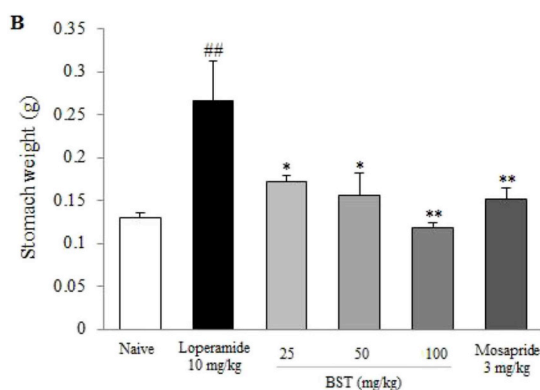
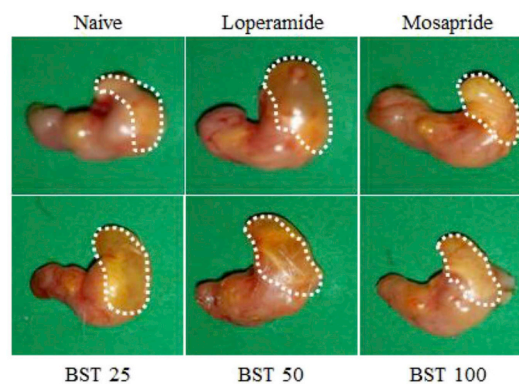
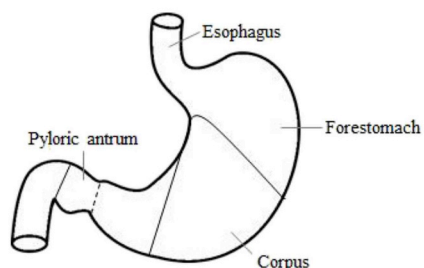


Fig. 2. Effects of BST on gastric emptying. For 3 days, the mice ($n = 8$ /each group) were orally administered BST (25, 50 and 100 mg/kg) or mosapride (3 mg/kg) and then injected with loperamide (10 mg/kg). After the administration of phenol red, visual observation (A), stomach weight (B) and assessment of gastric emptying (C) were recorded. The data are presented as the mean $\pm$ SEM. [#] $p < 0.05$, ^{##} $p < 0.01$ compared with the naive group; ^{*} $p < 0.05$, ^{**} $p < 0.01$ compared with the loperamide group. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

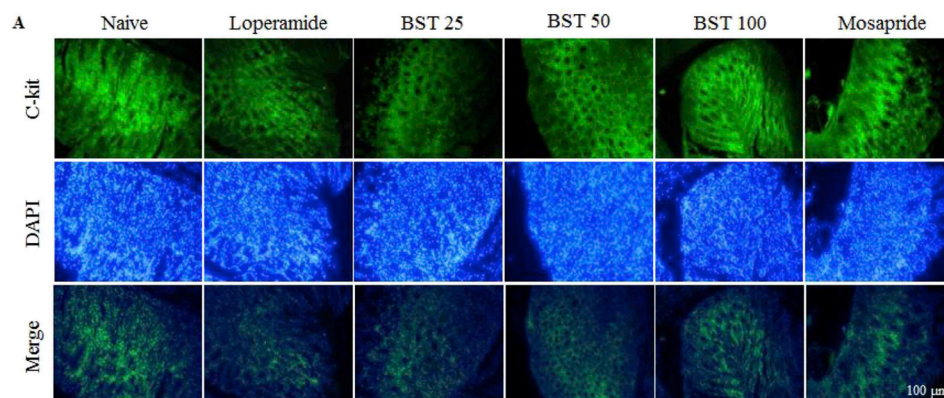
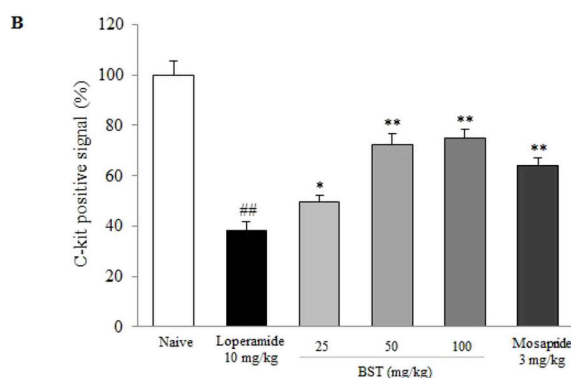


Fig. 3. Effects of BST on c-kit protein expression in stomach tissue. Immunofluorescence of c-kit in the stomach tissue (A) under an optical microscopy (200 $\times$ magnification) and its semi-quantification (B) was performed. Data are presented as the mean $\pm$ SD ($n = 3$). ^{##} $p < 0.01$ compared with the naive group; ^{*} $p < 0.05$, ^{**} $p < 0.01$ compared with the loperamide group.



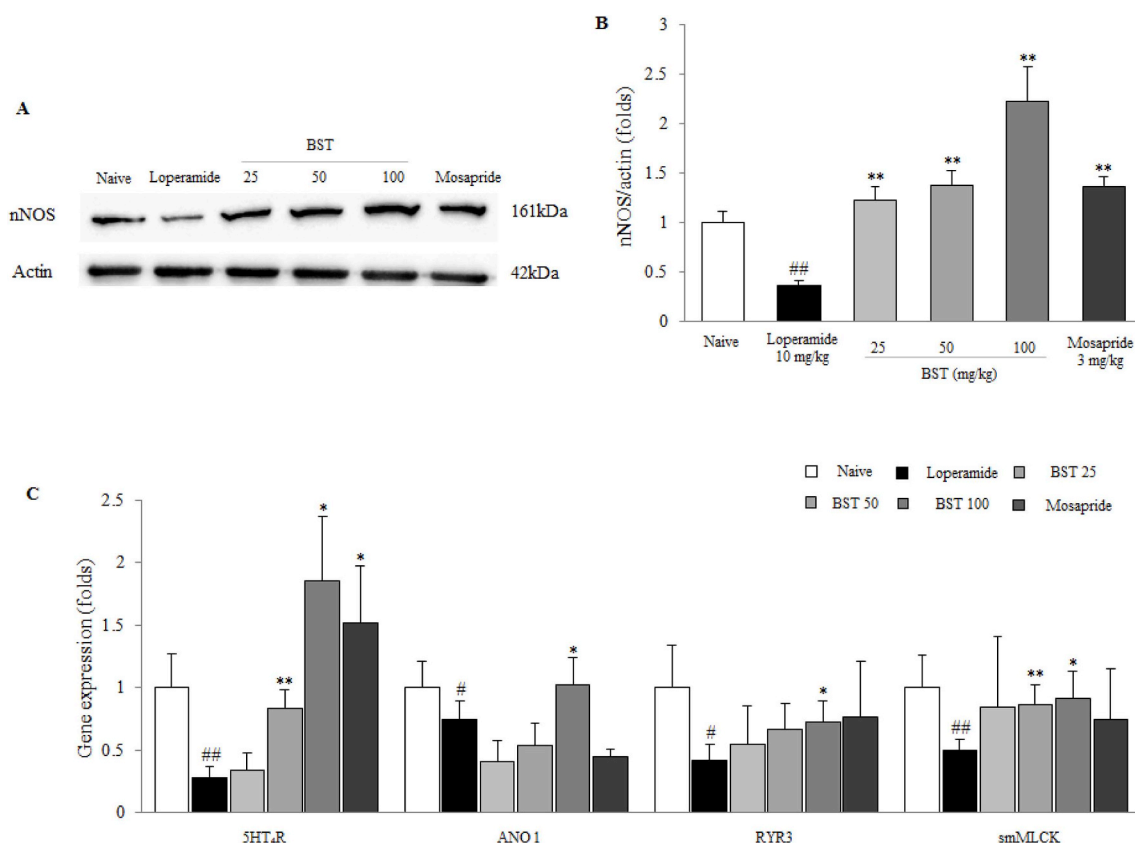


Fig. 4. Effects of BST on GI motility-associated molecules in stomach tissue. Western blot analysis for nNOS (A) and its semiquantification (B) was performed ($n = 3$). The mRNA expression analyses for four GI motility-associated genes were conducted (C) ($n = 3$) on the extracted stomach tissue. The data are presented as the mean $\pm$ SD. ## $p < 0.01$ compared with the naive group; * $p < 0.05$, ** $p < 0.01$ compared with the loperamide group. nNOS, neuronal nitric oxide synthase; 5HT₄R, 5HT₄ receptor; ANO1, anoctamin-1; RYR3, ryanodine receptor 3; smMLCK smooth muscle myosin light chain kinase.

induced delay of GI immobility was significantly tempered by the BST pretreatment (Fig. 5A and B).

In our experiments for exploring the mechanical actions of BST, such as histopathology, protein assays and RNA expressions, we had set up the separated mice groups which were not administrated with the phenol red solution or charcoal diet. The purpose was to examine the pharmacological activities of BST; those data would be a little different with the conditions accompanied with phenol red solution or charcoal diet. To explain the mechanisms responding therapeutic effects of BST, we next examined the c-kit expression level in the stomach tissue. The c-kit is a transmembrane receptor that produces the main signaling of ICC activation (Torihashi et al., 2004). The c-kit is mostly present in ICCs in GI track, and the suppressed c-kit level is commonly seen in tissue from the stomach or intestine of patients with FD or IBS (Huizinga et al., 1997; Lin et al., 1998). ICCs are pacemaker cells generating electrical slow-waves that transition to smooth muscle cells, leading to segmental and peristaltic contractions of the gastrointestinal track (Al-Shboul, 2013). As demonstrated in a clinical study, FD patients have electrogastric abnormalities due to slow-wave dysfunction (Lin and Chen, 2001). C-kit suppresses normal GI motility and disrupted intestinal slow-wave activity, resulting delayed gastric emptying (Klein et al., 2013). In our study, the BST pretreatment significantly attenuated the c-kit suppression in loperamide-injected stomach tissue (Fig. 3A and B). On the other hand, serotonin (5-hydroxytryptamine, 5-HT) and its counterpart 5-HT receptors (especially 5-HT₄R) take part in the regulation of smooth muscle motor activity and glandular secretion (Karamanolis and Tack, 2006; Penkova and Nikolova, 2017). In addition, 5-HT itself activates 5-HT₄R on sensory afferents to trigger the peristaltic reflex (Grider et al., 1998). Accordingly, 5-HT₄R agonists, such as mosapride that was used as the

positive control in the present study, are one of the main options for treatment of functional gastrointestinal disorder (Yang et al., 2017). The BST and mosapride pretreatment significantly restored the down-regulation of the 5-HT₄R gene by loperamide injection (Fig. 4C). Improvement of gastric emptying and GI motility of mosapride is attributed by 5-HT₄R activation. These therapeutic effects of mosapride are consistent with the findings of previous studies (Kawamura et al., 2012; Zhang et al., 2016). Also mosapride is typical medicine currently prescribed for FD patients (Curran and Robinson, 2008).

Our results from the protein level of nNOS and the four gene expressions (5-HT₄R, ANO1, RYR3 and smMLCK) in stomach tissue support the therapeutic effects of BST discussed above. As a representative neurotransmitter in the GI tract, nNOS-produced nitric oxide plays a critical role in the relaxation of smooth muscle cells (Terauchi et al., 2005). The nNOS^{-/-} mice have shown delayed gastric emptying and the nitric oxide synthase inhibitor, N^G-nitro-L-arginine methyl ester, delayed gastric emptying of a liquid meal in rats (Micci et al., 2005; Plourde et al., 1994). In humans, inhibition of nitric oxide synthase impairs accommodation and follows meal induced satiety (Tack et al., 2002). In addition, nNOS gene polymorphism is related to susceptibility to FD and to the pathologic condition of postprandial discomfort and epigastric pain (Park et al., 2014). Besides NO through cyclic GMP-dependent proteinase 1 β , phosphates the InsP3 receiver to reduce Ca²⁺ sensitivity thus reducing the activity of the pacemaker (Aoyama, 2004). The increase in Ca²⁺ concentration in ICC initiated by the ignition of RYR3 receptors activates Ca²⁺-sensitive chloride channels of the cell membrane to create spontaneous transient depolarizations (STDs), and is delivered to the gastrointestinal smooth muscles through the gap junction (Sanders et al., 2006). Activation of voltage-operated depolarization (VOD) by signals transmitted to smooth muscle membranes

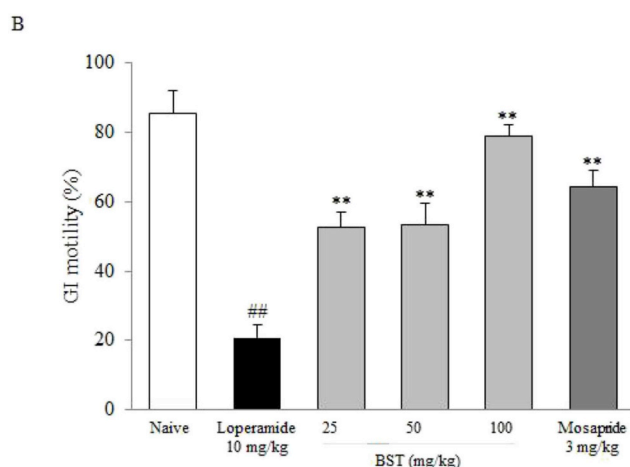
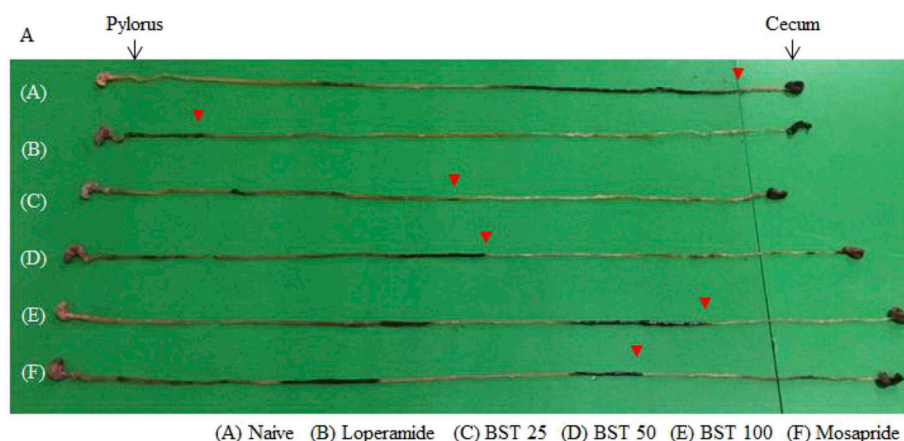


Fig. 5. Effects of BST on small intestinal motility. For 3 days, the mice ($n = 5$ /each group) were orally administered BST (25, 50 and 100 mg/kg) or mosapride (3 mg/kg) and then injected with loperamide (10 mg/kg). After 25 min of administration with charcoal, the distances covered were measured (A) and quantified (B). Red arrows indicate how much charcoal diet has moved into the cecum. The data are presented as the mean $\pm$ SD. $^{\#}p < 0.05$, $^{\#\#}p < 0.01$ compared with the naive group; $^*p < 0.05$, $^{**}p < 0.01$ compared with the loperamide group. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

increases Ca^{2+} in smooth muscle cells, combines with calmodulin to form a bond, and myosin slant formation by active MLC kinase causes smooth muscle contractions (Garcia et al., 1997). BST pretreatment significantly ameliorated the loperamide-induced suppression of the nNOS protein level, and smooth muscle contraction-related genes, 5-HT₄R, ANO1, RYR3 and smMLCK, were positively regulated by the BST pretreatment (Fig. 4A–C). Both of ICC activation and slow wave generation depend on the ANO1 function which leads to activate calcium efflux to intracellular ICC (Gomez-Pinilla et al., 2009; Sanders et al., 2010). Ca^{2+} spark makes slow wave, is regulated by both ANO1 at Cajal cell membrane and RYR3 molecule at endoplasmic reticulum respectively (Imaizumi et al., 1998; Liu et al., 2005). Reduction of smMLCK activity in smooth muscle results in gut dysmotility characterized by weak peristalsis (He et al., 2008). ANO1 and smMLCK were down-regulated in diabetic gastroparesis animal models (Hu and Feng, 2012; Mazzone et al., 2011). These results suggest that BST has pharmacological activity that regulates ICC in gastric tract organs. By regulating nNOS, BST may mediate normal peristalsis and activate the contraction-associated molecules.

BST is a traditional herbal formula, which was first recorded in a traditional medical textbook and is called *Shan-han-za-bing-lin* (傷寒雜病論) (World Health Organization, 2007). This drug is composed of seven medicinal herbs, including *Pinelliae tuber*, *Scutellariae radix*, and *Coptidis rhizome*, which have been used to treat conditions similar to dyspepsia (World Health Organization, 2007). Currently, BST is available commercially as a standardized herbal drug approved by MFDS in Korea. “Epigastric stuffiness (心下痞)”, the main traditional complaint treated by BST, is considered similar to the abdominal discomfort of functional dyspepsia (World Health Organization, 2007). To date, several animal and clinical studies have partially explained the effects

of BST on gastrointestinal disorders. A previous animal study suggested that BST increases important gut motility hormones, such as motilin, gastrin, and somatostatin (Naito et al., 2002). Two main compositional herbs of BST, *Zingiberis officinale* and *Pinelliae tuber*, have been reported to improve GI motility and delay gastric emptying, in an animal study as well as in a clinical trial (Lazzini et al., 2016; Naito et al., 2003). Therefore, BST may ameliorate the impaired smooth muscle motility in FD patients. Moreover, BST more effectively relieved early satiety than epigastric pain or burning (Park et al., 2013). These results indicate the pharmacological effects of BST, especially with respect to prokinetic action in GI motility. The dysmotility of GI track is well known to be linked to IBS as well as GERD (Evans et al., 1997; Pandolfino and Roman, 2011), thus our current results will be helpful to expand its clinical applications for other functional GI disorders.

In conclusion, our results are the first presentation of animal-based evidence for prokinetic activity of *Banha-sasim-tang*, and its underlying mechanisms may involve modulations of the interstitial cells of Cajal and the smooth muscle cells in stomach. The present study provided the scientific evidence for health benefits especially on patients with PDS-type of FD.

Conflicts of interest

The authors declare no conflicts of interest. All of the authors have approved the final article.

Author contribution statement

Yoo-Jin Jeon wrote the main manuscript text, and conducted experiments. Jin-Seok Lee conducted immunofluorescence staining and

performed a statistical analysis. Yong-Rae Cho treated mice with drugs and conducted ELSIA test. Sung-Bae Lee conducted western blotting. Won-Yong Kim prepared samples and conducted PCR. Seong-Soo Roh advised about the experiment schedules. Jin-Yong Joung advised about information of oriental medicine. Hwa-Dong Lee and Sung-Ok Moon analyzed the HPLC. Chang-Gue Son and Jung-Hyo Cho supervised the manuscript, and directed final version of all contents. All authors reviewed and approved this manuscript.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2019.111834>.

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